

Supplementary Material for When More Epochs Stop Helping: Coverage-Limited Scaling Laws for Sparse Replay

Anonymous submission

Notation and Assumptions

For coordinate $j \geq 1$,

$X_j = q_j Z_j E_j$, $Z_j \sim \text{Bernoulli}(p_j)$, $E_j \sim \text{Rad}$,
independently over coordinates and examples. The covariance is diagonal with $\lambda_j = p_j q_j^2$. Labels are $Y = \langle \theta, X \rangle$,
and $g_j = \lambda_j \theta_j^2$. Unless an exact constant is stated, there are constants $0 < c < C < \infty$ such that

$$\begin{aligned} c j^{-s} &\leq p_j \leq C j^{-s}, \\ c j^{-a} &\leq \lambda_j \leq C j^{-a}, \\ c j^{-b} &\leq g_j \leq C j^{-b}. \end{aligned}$$

We assume

$$s > 1, \quad a \geq s, \quad b > 1, \quad \sup_j p_j < 1.$$

For the coactive upper bound, we additionally assume

$$a - s + 1 < b < a + 1. \quad (1)$$

Define

$$\alpha = \frac{a - b + 1}{s}, \quad \beta = \frac{b - 1}{s}.$$

Then Eq. (1) is equivalent to $0 < \alpha < 1$, and $\alpha + \beta = a/s$.

Since $\sum_j p_j < \infty$, the Borel–Cantelli lemma implies that every input has finite support almost surely. For linear predictors,

$$\mathcal{R}(w) = \sum_j \lambda_j (w_j - \theta_j)^2.$$

For arbitrary measurable predictors, write $\mathcal{R}(f; \theta) = \mathbb{E}_X[(f(X) - \langle \theta, X \rangle)^2]$.

For N unique examples, define

$$\begin{aligned} \mathcal{S}_N &= \bigcup_{i=1}^N \text{supp}(X_i), & V_N &= |\mathcal{S}_N|, \\ A_N &= \sum_{j \in \mathcal{S}_N} \theta_j^2, & U_N &= \sum_{j \notin \mathcal{S}_N} g_j. \end{aligned}$$

The coordinate-presence indicators $I_{N,j} = \mathbf{1}\{j \in \mathcal{S}_N\}$ are independent across j , with

$$r_{N,j} := \mathbb{E} I_{N,j} = 1 - (1 - p_j)^N.$$

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A Power-Law Summation Lemma

The proofs repeatedly use the following regular-variation calculation.

Lemma 1 (Observed and missing power-law sums). *Let $p_j = c_p j^{-s}$ with $s > 1$ and $0 < c_p < 1$.*

1. *If $-1 < r < s - 1$ and $\xi = (r + 1)/s \in (0, 1)$, then*

$$\sum_{j \geq 1} j^r \{1 - (1 - p_j)^N\} \sim \frac{\Gamma(1 - \xi)}{r + 1} (c_p N)^\xi. \quad (2)$$

2. *If $b > 1$ and $\beta = (b - 1)/s$, then*

$$\sum_{j \geq 1} j^{-b} (1 - p_j)^N \sim \frac{1}{s} \Gamma(\beta) (c_p N)^{-\beta}. \quad (3)$$

The same expressions hold up to constant factors for two-sided power laws.

Proof. Set $J = (c_p N)^{1/s}$. For the observed sum, rescale $j = Ju$. On compact subsets of $(0, \infty)$,

$$1 - (1 - c_p j^{-s})^N \longrightarrow 1 - e^{-u^{-s}}.$$

The head and tail are dominated by $C \min\{1, u^{-s}\} u^r$, which is integrable exactly when $-1 < r < s - 1$. Therefore

$$\begin{aligned} J^{-(r+1)} \sum_j j^r \{1 - (1 - c_p j^{-s})^N\} \\ \longrightarrow \int_0^\infty u^r (1 - e^{-u^{-s}}) du. \end{aligned}$$

With $t = u^{-s}$,

$$\int_0^\infty u^r (1 - e^{-u^{-s}}) du = \frac{1}{s} \int_0^\infty t^{-\xi-1} (1 - e^{-t}) dt.$$

Integration by parts gives $\Gamma(1 - \xi)/(s\xi) = \Gamma(1 - \xi)/(r + 1)$.

For the missing sum, the same rescaling gives

$$J^{b-1} \sum_j j^{-b} (1 - c_p j^{-s})^N \longrightarrow \int_0^\infty u^{-b} e^{-u^{-s}} du.$$

The substitution $t = u^{-s}$ evaluates the integral as $s^{-1} \Gamma((b - 1)/s)$. Standard dyadic-shell bounds provide domination and also prove the two-sided-order statements. $\square$

Universal Coverage Lower Bound

Write $\theta_j = \tau_j \Omega_j$, where the magnitudes τ_j are fixed and the Ω_j are independent Rademacher signs.

Theorem 2 (Unseen-feature lower bound). *For every measurable, possibly randomized predictor trained on N unique examples,*

$$\mathbb{E}_{\Omega, D_N} \mathcal{R}(\hat{f}; \theta) \geq \sum_{j \geq 1} g_j (1 - p_j)^N.$$

Consequently, under the power-law assumptions,

$$\mathbb{E}_{\Omega, D_N} \mathcal{R}(\hat{f}; \theta) \gtrsim N^{-(b-1)/s}.$$

For exact $p_j = c_p j^{-s}$ and $g_j = c_g j^{-b}$, the lower bound is asymptotic to

$$\frac{c_g}{s} \Gamma\left(\frac{b-1}{s}\right) (c_p N)^{-(b-1)/s}.$$

Proof. Let $\mathcal{U}(D_N) = \{j : Z_{1j} = \dots = Z_{Nj} = 0\}$ be the unseen coordinates. Conditional on the training inputs, teacher signs outside $\mathcal{U}$, and the learner's internal randomness, no training label depends on $\{\Omega_j : j \in \mathcal{U}\}$. These signs remain independent and symmetric. For an independent test input, write

$$H(X) = \sum_{j \notin \mathcal{U}} \tau_j \Omega_j X_j.$$

For every finite $F \subset \mathcal{U}$, averaging squared error over the signs in F makes all cross terms vanish and yields

$$\begin{aligned} \mathbb{E}_{\Omega_F} \left(\hat{f}(X) - H(X) - \sum_{j \in \mathcal{U}} \tau_j \Omega_j X_j \right)^2 \\ \geq \sum_{j \in F} \tau_j^2 X_j^2. \end{aligned}$$

Let $F \uparrow \mathcal{U}$ and apply monotone convergence. Averaging over the test input gives at least $\sum_{j \in \mathcal{U}} g_j$. Finally, $\Pr(j \in \mathcal{U}) = (1 - p_j)^N$. The asymptotics follow from Lemma 1. $\square$

Because the left-hand side is a Bayes risk under the uniform sign prior, its value is no larger than the corresponding minimax risk. Hence the same rate is a minimax lower bound over deterministic teacher-sign choices.

Occupancy Calculus

Proposition 3 (Dictionary size, observed norm, and missing energy). *Under the power-law assumptions,*

$$\mathbb{E}V_N \asymp N^{1/s}, \quad \mathbb{E}A_N \asymp N^\alpha, \quad \mathbb{E}U_N \asymp N^{-\beta}.$$

For exact profiles,

$$\begin{aligned} \mathbb{E}V_N &\sim \Gamma(1 - 1/s) (c_p N)^{1/s}, \\ \mathbb{E}A_N &\sim \frac{c_g}{c_\lambda} \frac{\Gamma(1 - \alpha)}{a - b + 1} (c_p N)^\alpha, \\ \mathbb{E}U_N &\sim \frac{c_g}{s} \Gamma(\beta) (c_p N)^{-\beta}. \end{aligned}$$

Proof. Since $\mathbb{E}I_{N,j} = 1 - (1 - p_j)^N$,

$$\mathbb{E}V_N = \sum_j r_{N,j}.$$

Apply Eq. (2) with $r = 0$.

Next, $\theta_j^2 = g_j / \lambda_j \asymp j^{a-b}$. The source condition gives $-1 < a - b < s - 1$, so Eq. (2) with $r = a - b$ gives the claimed order and constant for $\mathbb{E}A_N$. Finally,

$$\mathbb{E}U_N = \sum_j g_j (1 - p_j)^N,$$

and Eq. (3) applies. $\square$

Lemma 4 (Capacity overflow). *There exists $c_m > 0$ such that $N \leq c_m m^s$ implies*

$$\Pr(V_N > m) \leq e^{-c_m}.$$

Proof. The $I_{N,j}$ are independent Bernoulli variables and $\mathbb{E}V_N \leq CN^{1/s}$. Choose c_m so that $N \leq c_m m^s$ implies $\mathbb{E}V_N \leq m/2$. A multiplicative Chernoff bound gives the result, first for a finite coordinate truncation and then by monotone convergence. $\square$

The plain-output theorem needs control of A_N on rare datasets.

Lemma 5 (Observed-norm moment). *For every $q > 1$ satisfying $q\alpha < 1$,*

$$(\mathbb{E}A_N^q)^{1/q} \leq C_q N^\alpha.$$

Such a q exists because $\alpha < 1$.

Proof. Let $r = a - b$, so $\alpha = (r + 1)/s$. By Minkowski's inequality,

$$\|A_N\|_{L_q} \leq \sum_j \theta_j^2 \|I_{N,j}\|_{L_q} = \sum_j \theta_j^2 r_{N,j}^{1/q}.$$

Set $J = N^{1/s}$. Since $r_{N,j} \lesssim \min\{1, Nj^{-s}\}$,

$$\|A_N\|_{L_q} \lesssim \sum_{j \leq J} j^r + N^{1/q} \sum_{j > J} j^{r-s/q}.$$

The second series converges because $r - s/q < -1$ is equivalent to $q\alpha < 1$. Both terms have order $J^{r+1} = N^\alpha$. $\square$

Exact One-Hot Replay

Suppose each example chooses a single coordinate $J_i \sim (p_j)$ and equals $X_i = E_i q_{J_i} e_{J_i}$. Let $C_j = \sum_{i=1}^n \mathbf{1}\{J_i = j\}$.

Proposition 6 (Exact finite-epoch risk). *Train a width- m model from zero by squared-loss SGD with step η for K repeated passes over the same dataset. Then*

$$\mathcal{R}_D = \sum_{j \leq m} g_j (1 - \eta q_j^2)^{2KC_j} + \sum_{j > m} g_j,$$

and

$$\mathbb{E}_D \mathcal{R}_D = \sum_{j \leq m} g_j [1 - p_j + p_j (1 - \eta q_j^2)^{2K}]^n + \sum_{j > m} g_j.$$

Proof. For the residual $\delta_j = w_j - \theta_j$, a sample on coordinate j gives $\delta_j^+ = (1 - \eta q_j^2)\delta_j$. Coordinate updates commute, so after K passes coordinate j receives $K C_j$ multiplications. The expectation follows from the binomial probability-generating function. $\square$

Theorem 7 (One-hot three-frontier law). *Suppose $\sup_j p_j < 1$ and $0 < \eta q_j^2 \leq \rho < 1$. Then*

$$\mathbb{E}_D \mathcal{R}_D \asymp J_*^{1-b}, \quad J_* = \min\{m, (\eta n K)^{1/a}, n^{1/s}\}.$$

Equivalently,

$$\mathbb{E}_D \mathcal{R}_D \asymp m^{-(b-1)} + (\eta n K)^{-(b-1)/a} + n^{-(b-1)/s}.$$

Proof. Put $u_j = \eta q_j^2$ and $r_j = 1 - (1 - u_j)^{2K}$. Uniformly for $0 < u_j \leq \rho$,

$$r_j \asymp \min\{1, K u_j\}.$$

Therefore

$$np_j r_j \asymp \min\{np_j, \eta n K \lambda_j\} \asymp \min\{n j^{-s}, \eta n K j^{-a}\}.$$

Because $p_j r_j \leq \sup_j p_j < 1$, the exact factor $(1 - p_j r_j)^n$ is bounded above and below by exponentials with constant multiples of $np_j r_j$ in the exponent. It remains to estimate

$$S = \sum_{j \leq m} j^{-b} \exp\{-c \min(A j^{-a}, B j^{-s})\} + \sum_{j > m} j^{-b},$$

where $A = \eta n K$ and $B = n$. Let $J = \min\{m, A^{1/a}, B^{1/s}\}$. For $j \leq J$, since $a \geq s$,

$$\min\{A j^{-a}, B j^{-s}\} \geq (J/j)^s.$$

A dyadic-shell bound gives an $O(J^{1-b})$ contribution from the represented core, while all indices above J contribute $O(J^{1-b})$. For the lower bound, if $J = m$, the omitted tail is $\Omega(J^{1-b})$. Otherwise, either $m < 2J$ and the omitted tail again suffices, or the represented shell $[J, 2J]$ has a bounded exponent and contributes $\Omega(J^{1-b})$. $\square$

Coactive Clipped Replay

For a fixed dataset D_N , the sparse dictionary stores $\mathcal{S}_N$ when $V_N \leq m$. On capacity overflow, the capacity-guarded algorithm outputs zero. Otherwise, at update t it samples I_t uniformly from $\{1, \dots, N\}$ and uses

$$\gamma_i = \min\left\{\eta, \frac{\rho}{\|X_i\|^2}\right\},$$

$$w_{t+1} = w_t - \gamma_{I_t} X_{I_t} (X_{I_t}^\top w_t - Y_{I_t}).$$

where the second argument is $+\infty$ for a zero row. Assume $\eta \sup_j q_j^2 \leq \rho \leq 1$ and define $\bar{w}_T = T^{-1} \sum_{t=0}^{T-1} w_t$.

Let $C_{N,j}$ count examples with support exactly $\{j\}$. Since $\sum_j p_j < \infty$,

$$\pi_0 = \prod_{k \geq 1} (1 - p_k) > 0,$$

$$\nu_j := \Pr\{\text{supp}(X) = \{j\}\} = \frac{\pi_0 p_j}{1 - p_j} \asymp p_j.$$

Fix $0 < \kappa \leq \pi_0/2$ and define

$$\mathcal{A}_N = \{j : C_{N,j} \geq \kappa N p_j\}, \quad \hat{w}_T = P_{\mathcal{A}_N} \bar{w}_T.$$

Lemma 8 (Pathwise contraction). *Let $\delta_t = w_t - \theta_{\mathcal{S}_N}$. Every update satisfies*

$$\|\delta_{t+1}\|^2 \leq \|\delta_t\|^2 - \gamma_{I_t} (X_{I_t}^\top \delta_t)^2.$$

Hence $\|\delta_t\| \leq \sqrt{A_N}$ along every replay path.

Proof. For a row x , $\delta^+ = (I - \gamma x x^\top) \delta$, and therefore

$$\|\delta^+\|^2 = \|\delta\|^2 - \gamma(2 - \gamma\|x\|^2)(x^\top \delta)^2.$$

The clipping rule ensures $\gamma\|x\|^2 \leq \rho \leq 1$. $\square$

Lemma 9 (Singleton-weighted energy inequality). *Conditioned on any fixed dataset with $V_N \leq m$,*

$$\frac{\eta}{N} \sum_{t=0}^{T-1} \mathbb{E}_{\text{sgd}} \sum_j C_{N,j} q_j^2 \delta_{t,j}^2 \leq A_N. \quad (4)$$

Proof. Average Lemma 8 over the next uniformly sampled row. All terms are nonnegative, so retain only singleton rows. A singleton- j row is $\pm q_j e_j$ and is not clipped because $\eta q_j^2 \leq \rho$. Thus

$$\mathbb{E}_t \|\delta_{t+1}\|^2 \leq \|\delta_t\|^2 - \frac{\eta}{N} \sum_j C_{N,j} q_j^2 \delta_{t,j}^2.$$

Telescope and use $\|\delta_0\|^2 = A_N$. $\square$

Lemma 10 (Energy outside the anchor set). *There is $c > 0$ such that*

$$\mathbb{E} \sum_{j \notin \mathcal{A}_N} g_j \leq C N^{-\beta}.$$

Proof. For fixed j , $C_{N,j} \sim \text{Binomial}(N, \nu_j)$ with $\nu_j \geq \pi_0 p_j$. Since $\kappa \leq \pi_0/2$, the anchor threshold is at most half the mean. A multiplicative Chernoff bound gives

$$\Pr(j \notin \mathcal{A}_N) \leq e^{-c N p_j}.$$

Therefore

$$\mathbb{E} \sum_{j \notin \mathcal{A}_N} g_j \leq C \sum_j j^{-b} e^{-c N j^{-s}} \asymp N^{-\beta}$$

by the same rescaling as Lemma 1. $\square$

Theorem 11 (Sharp anchor-masked coactive replay). *Suppose $N \leq c_m m^s$. The capacity-guarded, anchor-masked output satisfies*

$$\mathbb{E} \mathcal{R}(\hat{w}_T) \leq C \left[\frac{N^\alpha}{\eta T} + N^{-\beta} + e^{-cm} \right].$$

Proof. Condition on a fixed dataset with $V_N \leq m$. For $j \in \mathcal{A}_N$, $C_{N,j} \geq \kappa N p_j$, so Eq. (4) implies

$$\sum_{t < T} \mathbb{E}_{\text{sgd}} \sum_{j \in \mathcal{A}_N} \lambda_j \delta_{t,j}^2 \leq \frac{A_N}{\kappa \eta}.$$

By Jensen,

$$\mathbb{E}_{\text{sgd}} \sum_{j \in \mathcal{A}_N} \lambda_j (\bar{w}_{T,j} - \theta_j)^2 \leq \frac{A_N}{\kappa \eta T}.$$

The masked coordinates contribute exactly $\sum_{j \notin \mathcal{A}_N} g_j$. Average over the dataset, apply Proposition 3 and Lemma 10, and add the capacity-overflow contribution using Lemma 4. The zero predictor has finite risk $\sum_j g_j$. $\square$

We now remove the anchor mask. The analysis uses simultaneous singleton coverage only as a proof event; the algorithm itself is unchanged and never aborts because of singleton counts.

Lemma 12 (High-coverage core). *For every $R > 0$, there is a sufficiently small $c_J > 0$ such that, with*

$$J_N = \left\lfloor c_J \left(\frac{N}{\log(eN)} \right)^{1/s} \right\rfloor,$$

the event

$$\mathcal{G}_N = \{C_{N,j} \geq \kappa N p_j \text{ for every } j \leq J_N\}$$

satisfies $\Pr(\mathcal{G}_N^c) \leq N^{-R}$.

Proof. For $j \leq J_N$, the mean singleton count obeys $N\nu_j \geq c_N J_N^{-s} \geq cc_J^{-s} \log(eN)$. Chernoff gives $\Pr(C_{N,j} < \kappa N p_j) \leq N^{-R-2}$ after decreasing c_J , and a union bound over at most N coordinates completes the proof. $\square$

Theorem 13 (Plain Polyak average). *Suppose $N \leq c_m m^s$. The capacity-guarded plain Polyak average, with no anchor mask and no singleton-support abort, satisfies*

$$\mathbb{E}\mathcal{R}(\bar{w}_T) \leq C \left[\frac{N^\alpha}{\eta T} + \{\log(eN)\}^{a/s} N^{-\beta} + e^{-cm} \right].$$

Proof. Condition on $V_N \leq m$. On $\mathcal{G}_N$, repeat the proof of Theorem 11 on coordinates $j \leq J_N$ to obtain core risk $A_N/(\kappa\eta T)$. For represented coordinates above J_N , path-wise contraction gives $\|\bar{\delta}_T\|^2 \leq A_N$, so monotonicity of the spectrum gives

$$\sum_{\substack{j > J_N \\ j \in \mathcal{S}_N}} \lambda_j \bar{\delta}_{T,j}^2 \leq C \lambda_{J_N+1} A_N.$$

The unobserved coordinates contribute U_N . Hence, on $\mathcal{G}_N$,

$$\mathbb{E}_{\text{sgd}} \mathcal{R}(\bar{w}_T) \leq \frac{A_N}{\kappa\eta T} + C \lambda_{J_N+1} A_N + U_N.$$

After expectation,

$$\begin{aligned} \lambda_{J_N+1} \mathbb{E} A_N &\lesssim \left(\frac{N}{\log(eN)} \right)^{-a/s} N^\alpha \\ &= \{\log(eN)\}^{a/s} N^{-\beta}. \end{aligned}$$

On $\mathcal{G}_N^c$, Lemma 8 implies $\mathcal{R}(\bar{w}_T) \leq C(A_N + 1)$. Choose $q > 1$ with $q\alpha < 1$. By Lemma 5 and Hölder,

$$\mathbb{E}[(A_N + 1)\mathbf{1}_{\mathcal{G}_N^c}] \leq C(N^\alpha + 1) \Pr(\mathcal{G}_N^c)^{1-1/q}.$$

Lemma 12 permits an arbitrarily large polynomial exponent R , so this term is $O(N^{-\beta})$. Finally add the capacity-overflow probability from Lemma 4. $\square$

Corollary 14 (All-data saturation). *Set $N = n$, $T = nK$, and $m \gtrsim n^{1/s}$. Then*

$$\mathbb{E}\mathcal{R}(\hat{w}_{nK}) \lesssim \frac{n^{\alpha-1}}{\eta K} + n^{-\beta}.$$

Therefore $K = O(\eta^{-1} n^{a/s-1})$ suffices to reach the coverage floor. The plain average has the same exponent with the logarithmic factor from Theorem 13.

Proof. Balance the two terms. Since $\alpha + \beta = a/s$,

$$\frac{n^{\alpha-1}}{\eta K} \lesssim n^{-\beta} \iff K \gtrsim \eta^{-1} n^{\alpha+\beta-1} = \eta^{-1} n^{a/s-1}.$$

$\square$

Corollary 15 (Frontier-matched schedule). *Let*

$$J_\star = \min\{m, n^{1/s}, (\eta T)^{1/a}\}$$

and choose

$$N_\star = \left\lfloor c \min\{n, m^s, (\eta T)^{s/a}\} \right\rfloor.$$

For $J_\star$ larger than a fixed constant,

$$\mathbb{E}\mathcal{R}(\hat{w}_T) \leq C J_\star^{1-b}.$$

Equivalently,

$$\mathbb{E}\mathcal{R}(\hat{w}_T) = O\left(m^{-(b-1)} + (\eta T)^{-(b-1)/a} + n^{-(b-1)/s}\right).$$

Proof. The choice of $N_\star$ guarantees both $N_\star \lesssim m^s$ and $\eta T \gtrsim N_\star^{a/s}$. Therefore

$$\frac{N_\star^\alpha}{\eta T} \lesssim N_\star^{\alpha-a/s} = N_\star^{-\beta}.$$

Moreover $N_\star^{1/s} \asymp J_\star$, so $N_\star^{-\beta} = J_\star^{1-b}$. Since $1-b < 0$, the largest of the three frontier terms is equivalent within a factor three to their sum. $\square$

Budgeted Acquisition and Replay

The main paper states a budgeted version of the frontier-matched schedule. We record the argument explicitly. Let c_d be the cost of acquiring one unique example, c_u the cost of one update, and let $c_d N + c_u T \leq B$. Define

$$N_B = \left\lfloor c_0 \min\left\{n, m^s, (\eta B/c_u)^{s/a}, B/c_d\right\} \right\rfloor$$

for a sufficiently small numerical c_0 , and set $T_B = \lfloor (B - c_d N_B)/c_u \rfloor$.

Corollary 16 (Budgeted frontier matching). *For N_B above a fixed constant,*

$$\mathbb{E}\mathcal{R}(\hat{w}_{T_B}) = O\left(m^{-(b-1)} + n^{-(b-1)/s} + (\eta B/c_u)^{-(b-1)/a} + (B/c_d)^{-(b-1)/s} + e^{-cm}\right).$$

Proof. Choose $c_0 \leq 1/2$. Since $N_B \leq c_0 B/c_d$, $T_B \geq B/(2c_u) - 1$, and hence $T_B \asymp B/c_u$. Moreover,

$$N_B^{1/s} \lesssim \min\left\{n^{1/s}, m, (\eta B/c_u)^{1/a}, (B/c_d)^{1/s}\right\}.$$

Apply Corollary 15 with $T = T_B$ and the selected subset size N_B . The optimization contribution obeys $N_B^\alpha/(\eta T_B) \lesssim N_B^{-\beta}$ because $N_B \lesssim (\eta B/c_u)^{s/a}$ and $\alpha + \beta = a/s$. Converting $N_B^{-\beta}$ to the maximum of the four frontier tails gives the stated sum up to a constant factor. $\square$

When n and m are nonbinding and $a > s$, the acquisition constraint $N_B \leq B/c_d$ is asymptotically inactive, so $N_B \asymp B^{s/a}$, $T_B \asymp B$, and $K_B = T_B/N_B \asymp B^{1-s/a}$. For $a = s$, fresh data and updates must instead receive constant fractions of the budget. In either case the allocation exponent is independent of b , for the same reason as the saturation epoch: b converts learned rank into risk but does not determine the rank at which optimization and coverage meet.

Frontier Elasticities

For completeness, let

$$L = C_m m^{-\delta} + C_T (\eta T)^{-\gamma} + C_N N^{-\beta}.$$

Direct differentiation gives

$$-\frac{\partial \log L}{\partial \log m} = \delta \frac{C_m m^{-\delta}}{L}, \quad -\frac{\partial \log L}{\partial \log T} = \gamma \frac{C_T (\eta T)^{-\gamma}}{L},$$

and the analogous identity for N . Dividing by the respective exponents and summing gives the simplex identity stated in the main paper. Centered finite differences on a logarithmic grid consistently estimate these elasticities whenever the scaling proxy is locally accurate. The identity also extends immediately to additional additive bottlenecks: the normalized local elasticities form a simplex whose coordinates are the corresponding fractions of predicted risk.

Input-Only Saturation Estimation

For exact profiles,

$$p_j = c_p j^{-s}, \quad \lambda_j = c_\lambda j^{-a}.$$

Eliminating j gives

$$q_j^2 = \frac{\lambda_j}{p_j} = c_\lambda c_p^{-a/s} p_j^{a/s-1}. \quad (5)$$

At the coverage rank $p_j \asymp 1/n$. A feature near this rank appears a constant number of times per epoch and requires $K \eta q_j^2 \asymp 1$ to be optimized. Consequently,

$$K_{\text{sat}} \asymp \frac{1}{\eta q^2(p=1/n)} = \eta^{-1} c_\lambda^{-1} c_p^{a/s} n^{a/s-1}.$$

The empirical estimator uses only the design matrix. Let M_j be the number of occurrences of feature j and let

$$\hat{p}_j = M_j/n, \quad \hat{q}_j^2 = \frac{1}{M_j} \sum_{i: Z_{ij}=1} X_{ij}^2.$$

For coordinates with $M_j \geq r_{\min}$, fit

$$\log \hat{q}_j^2 = \hat{c} + \hat{r} \log \hat{p}_j$$

and output

$$\hat{K}_{\text{sat}} = \eta^{-1} e^{-\hat{c}} n^{\hat{r}}.$$

The experiments use $r_{\min} = 8$. This threshold trades the number of fitted coordinates against small-count noise. A uniform plug-in theorem for an empirical anchor mask or for the stopping estimator would require additional control of feature selection and is not claimed here.

Experimental Details

Sparse design generation

For ambient dimension d , the experiments use

$$p_j = \frac{\mu}{\zeta(s)} j^{-s}, \quad \lambda_j = j^{-a}, \\ q_j^2 = \frac{\lambda_j}{p_j}, \quad g_j = j^{-b}.$$

For each coordinate, the code first draws its $\text{Binomial}(n, p_j)$ occupancy and then chooses that many rows uniformly without replacement. This is exactly distributionally equivalent to drawing all nd independent Bernoulli masks, while requiring $O(d + \text{nnz})$ memory. Rademacher signs are then assigned to nonzero entries. The teacher is $\theta_j = \pm \sqrt{g_j/\lambda_j}$ with independent signs.

Population risk in the first d coordinates is computed exactly from the coefficient error. Every reported value adds the analytic target tail $\zeta(b, d+1)$. The exact unseen-feature floor is

$$\sum_{j \leq d} g_j (1 - p_j)^n + \zeta(b, d+1).$$

Optimization protocols

The primary protocol samples stored rows with replacement and uses

$$\gamma_i = \min\{\eta, 0.9/\|X_i\|^2\}.$$

We record both the last iterate and the uniform average of pre-update iterates. Cyclic and independently reshuffled protocols are robustness checks. No support-abort event is used in any experiment. The anchor experiment masks only the final averaged vector, with $\kappa = \pi_0/2$ and $\pi_0 = \prod_{j \leq d} (1 - p_j)$.

Primary and robustness sweeps

The primary sweep uses $(s, a, b, \mu, \eta) = (1.5, 2, 2, 1, 0.2)$, $d = 4096$, $n \in \{256, 512, \dots, 8192\}$, and $K \in \{1, 2, \dots, 512\}$ on a logarithmic grid, with five seeds. The fitted fixed-exponent law for the averaged iterate is

$$0.916(\eta n K)^{-1/2} + 1.125n^{-2/3},$$

with $R^2 = 0.9805$ and median relative error 0.0954. The corresponding last-iterate values are $R^2 = 0.9759$ and median relative error 0.0997.

The progressive sweep uses target updates $T = 2^{12}, \dots, 2^{20}$, $N = \lfloor (\eta T)^{s/a} \rfloor$, and $K = \lfloor T/N \rfloor$. The measured slopes are -0.5016 for the average and -0.4970 for the last iterate. The fixed-compute sweep holds $T = 2^{18}$ and varies $N \in \{256, 512, \dots, 32768\}$. The last-iterate risk falls from 0.04302 to 0.00335, a factor of 12.84.

The source sweep holds $(a, s) = (2, 1.5)$ and tests $b \in \{1.4, 1.7, 2.0, 2.4\}$. The measured coverage slopes are $(-0.2633, -0.4580, -0.6532, -0.9150)$, compared with $(-0.2667, -0.4667, -0.6667, -0.9333)$. The spectrum grid additionally tests $(a, s, b) = (2, 1.5, 2), (2.4, 1.5, 2), (2.5, 2, 2.5)$; fixed-exponent two-term fits have $R^2 = (0.989, 0.996, 0.993)$.

The clipping stress test sweeps $\mu \in \{1, 1.75, 2.25\}$ and $\eta \in \{0.2, 0.4, 0.8, 1.2\}$. At $(\mu, \eta) = (2.25, 1.2)$, unclipped reshuffling has mean risk 1.16×10^{85} by epoch four and later overflows. Clipping reaches 0.00708 by epoch 64; the coverage floor is 0.00618.

Capacity sweep

To isolate the model frontier, we fix $n = 32768$, $K = 256$, $(s, a, b) = (1.5, 2, 2)$, and vary $m \in \{64, 128, 256, 512, 1024, 2048, 4096\}$. The generator treats m as the representable ambient rank and the risk adds the

Table 1: Input-only saturation diagnostics. “True r ” is $a/s - 1$; the factor column is the median of $\max\{\hat{K}/K, K/\hat{K}\}$.

Configuration	true r	mean $\hat{r}$	factor error
(2, 1.5, 2)	0.333	0.349	1.68
(2.4, 1.5, 2)	0.600	0.629	1.49
(2.5, 2, 2.5)	0.250	0.257	1.55

exact tail beyond m . Across five seeds, fitting the first three widths gives average-risk slope -0.9911 ; the exact target-tail slope over the same points is -0.9958 . The transition to coverage-limited behavior occurs around $m = 1024$.

Stopping-rule sweep

We test three configurations:

$$(a, s, b) \in \{(2, 1.5, 2), (2.4, 1.5, 2), (2.5, 2, 2.5)\},$$

with $n \in \{256, 512, 1024, 2048, 4096, 8192\}$, $d = 8192$, and five seeds. The measured saturation is the first log-interpolated epoch for which averaged risk is at most 1.25 times the exact coverage floor. Features must appear at least eight times to enter the input-only regression.

The complete release now contains 795 independent training runs and 5,400 recorded population-risk checkpoints.

Reproducibility

The repository contains:

- `experiments/exact_pilots.py`, evaluating the exact one-hot and singleton-filtered formulas;
- `experiments/multi_active_replay.py`, generating the primary multi-active sweeps;
- `experiments/frontier_diagnostics.py`, running the width and input-only stopping experiments;
- `experiments/plot_results.py`, regenerating all paper figures;
- `experiments/validate_results.py`, checking result schemas, counts, fitted slopes, and reported summary values.

The principal commands are `pip install -r requirements.txt`, `make full`, running `experiments/frontier_diagnostics.py`, and then `make validate` and `make figures`. All random seeds, dimensions, spectral exponents, epoch grids, and fitting windows are specified in the scripts and checked-in CSV files.